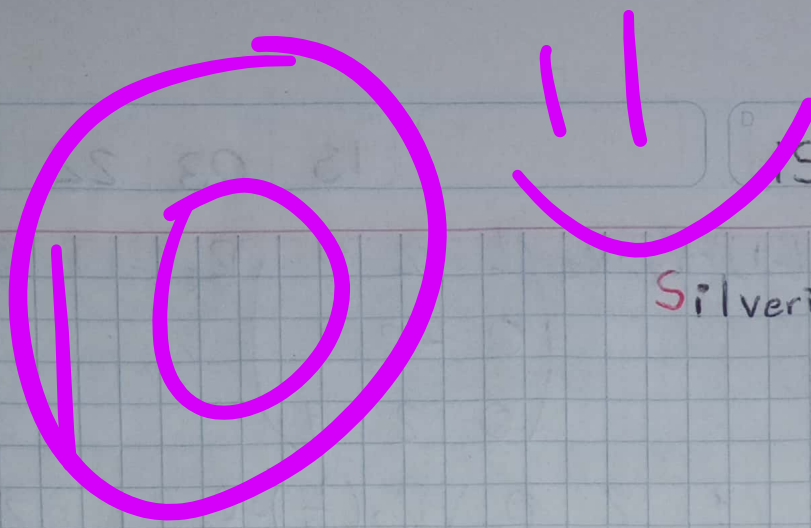




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Álgebra Lineal
Grupo 17
Semestre 2022-2

Tarea 9

Sean el espacio vectorial real $P = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$ y la transformación lineal $F: P \rightarrow P$ cuya regla de correspondencia es:

$$F(ax^2 + bx + c) = (a + 2b)x^2 + (a + c)x + (2b - c)$$

- a) Determinar el núcleo de F y la dimensión de $N(F)$.
- b) El recorrido de F .

Solución

a) $N(F)$

$$N(F) = \{v \in V \mid F(v) = \vec{0}_W; \vec{0}_W \in W\}$$

$$N(F) = \{(ax^2 + bx + c) \in P \mid F(a, b, c) = (0, 0, 0); (0, 0, 0) \in P\}$$

$\begin{matrix} \text{2º res} & & \text{1º res} \end{matrix}$

$$1^\circ F(a, b, c) = (0, 0, 0)$$

$$(a + 2b)x^2 + (a + c)x + (2b - c) = (0, 0, 0)$$

$$\begin{cases} a + 2b = 0 \\ a + c = 0 \\ 2b - c = 0 \end{cases} \text{ sistema de ecuaciones}$$

$$\begin{aligned} a &= -c \\ b &= -\frac{c}{2} \\ c &= -2\left(-\frac{c}{2}\right) \\ c &= c \end{aligned}$$

$$a = -c, \quad b = 1 - \frac{a}{2} = 1 - \frac{-c}{2} = 1 + \frac{c}{2}, \quad c = -2\left(\frac{b}{2}\right) = -2\left(\frac{1 + \frac{c}{2}}{2}\right) = -2\left(\frac{1}{2} + \frac{c}{4}\right) = -1 - \frac{c}{2}$$

$$c = 0$$

$$b = -\frac{1}{2}c$$

$$c = -2\left(-\frac{1}{2}c\right)$$

$$c = c$$

$$N(T) = \left\{ \left(1 - c, \frac{1}{2}c, c\right) \mid c \in \mathbb{R} \right\}$$

$$N(T) = \left\{ -cx^2 + \frac{1}{2}cx + c \mid c \in \mathbb{R} \right\}$$

$$\dim N(T) = 1$$

$$N(T) = \{ (ax^2 + bx + c) \in \mathbb{R} \}$$

$$N(T) = \left\{ \left(\frac{1}{2}c x^2 + (1 - \frac{c}{2})x + c \right) \mid c \in \mathbb{R} \right\}$$

$$\text{Base } N(T) = \left\{ \left(\frac{1}{2}x^2 + \left(1 - \frac{1}{2}\right)x + 1 \right) \right\}$$

$$\dim N(T) = 1$$

b) Recorrido de F

$$TM(T) = \left\{ (ax^2 + bx + c) \in \mathbb{R} \mid T(a_2x^2 + b_2x + c_2) = \begin{pmatrix} 0x^2 + 0x + 0 \\ 0x^2 + 0x + 0 \end{pmatrix} \in \mathbb{R} \right\}$$

$$T(ax^2 + bx + c) = (a_2x^2 + b_2x + c_2)$$

$$T[(a+2b)x^2 + (a+c)x + (2b-c)] = (a_2x^2 + b_2x + c_2)$$

$$\begin{cases} a+2b=a_2 \\ a+c=b_2 \\ 2b-c=c_2 \end{cases} \begin{matrix} \text{sis.} \\ \text{de} \\ \text{ecu.} \end{matrix}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a_2 \\ 1 & 0 & 1 & b_2 \\ 0 & 2 & -1 & c_2 \end{array} \right) R_1 - R_2 = R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a_2 \\ 0 & -2 & 1 & -a_2+b_2 \\ 0 & 2 & -1 & c_2 \end{array} \right) R_2 \leftrightarrow R_3$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a_2 \\ 0 & 2 & -1 & c_2 \\ 0 & -2 & 1 & -a_2+b_2 \end{array} \right) R_2 + R_3 = R_3$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & a_2 \\ 0 & 2 & -1 & -a_2+b_2 \\ 0 & 0 & 0 & -a_2+b_2+c_2 \end{array} \right)$$

$$-a_2+b_2+c_2$$

$$a_2 = a_2 //$$

$$c_2 = c_2 //$$

$$b = a_2 - c_2 //$$

$$IM(T) = \{(a_2x^2 + b_2x + c_2) \in P\} \quad TM(T) = \{(a_2x^2 + (a_2 - c_2)x + c_2) \mid a_2, c_2 \in \mathbb{R}\}$$

Sea el espacio vectorial real $P = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$ y la transformación lineal $T: \mathbb{R}^2 \rightarrow P$ tal que:

$$T(1, 2) = 2x^2 - x + 1$$

$$T(2, 1) = x^2 + x + 2$$

Determinar la regla de correspondencia de T .

Solución

$$(a, b) = \alpha_1(1, 2) + \alpha_2(2, 1)$$

$$(a, b) = (\alpha_1 + 2\alpha_2, 2\alpha_1 + \alpha_2)$$

$$\alpha_1 + 2\alpha_2 = a \quad \begin{matrix} \text{sis.} \\ \text{de} \\ \text{ecu.} \end{matrix}$$

$$2\alpha_1 + \alpha_2 = b$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 2 & 1 & b \end{array} \right) R_2 - 2R_1 = R_2 \quad \left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -3 & 2a-b \end{array} \right) R_2 \left(\frac{1}{-3} \right) = R_2$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & 1 & \frac{2a-b}{3} \end{array} \right) R_2(2) - R_2 = R_1 \quad \left(\begin{array}{cc|c} 1 & 0 & -\frac{a+2b}{3} \\ 0 & 1 & \frac{2a-b}{3} \end{array} \right)$$

$$\alpha_1 = \frac{-a+2b}{3}$$

$$\alpha_2 = \frac{2a-b}{3}$$

$$T(a, b) = \alpha_1 T(1, 2) + \alpha_2 T(2, 1)$$

$$T(a, b) = \left(\frac{-a+2b}{3} \right) (2x^2 - x + 1) + \left(\frac{2a-b}{3} \right) (x^2 + x + 2)$$

$$T(a, b) = \left(\frac{-2a+4b+2a-b}{3} \right) x^2 + \left(\frac{a-2b+2a-b}{3} \right) x +$$

$$\left(\frac{-a+2b+4a-2b}{3} \right)$$

$$T(a, b) = bx^2 + (a-b)x + a$$

$$T(1, 2) = 2$$

$$T(2, 1) = 2$$

$$T(1, 2) = \frac{(2(1))^2 - 4(1)^2(2) + 2(1)(4) - (1)^2 - 2(2) + (1)}{3}$$

$$T(1, 2) = \frac{2 - 8 + 4 - 1 + 3}{3}$$

$$(1, 2) = \frac{-6}{3}$$

$$2$$